

Real people meeting virtually real people — a review of some experiments in shared virtual environments

M Slater

This paper reports on experimental studies looking into the experience of people who are collaborating in shared virtual environments. In these environments, people are represented as ‘avatars’ in virtual space. The studies reported here suggest that the way in which people are represented has a strong effect on others’ perceptions. The effects on the feeling of closeness or ‘accord’ of people who meet in this way are contrasted with more natural real-life meetings where all the normal social cues are available. The characteristics of the systems used to access the shared virtual environment are considered. Different systems offering differing degrees of perceived ‘immersion’ in an environment are discussed, and their effects on social behaviours such as leadership are considered.

1. Introduction

The BBC TV series ‘Dr Who’ popularised the race of beings known as the ‘Daleks’¹. A Dalek (see Fig 1) is completely encased in a metallic shell, through which it can move through the world on rollers, only over certain types of relatively flat, and even surfaces. It has only one limb, which is used as a weapon, and occasionally as an end-effector for the manipulation of objects, together with an eye-piece on a stalk delivering a relatively small field of view. There is no direct evidence about Dalek’s auditory capabilities — however, this is unlikely to be good, since Daleks have a habit of repeating most things they say several times. Daleks tend to shout rather than talk, another indication of poor auditory capability and a lossy information channel. This evolutionary development was the result of a war-induced nuclear holocaust, thousands of years in the past.

Daleks experience the world in only one particular way — as Daleks. They typically cannot survive outside their metal casing. Humans on the other hand have a very different evolutionary history, have relatively flexible bodies, including hands capable of very precise manipulations and gross manoeuvres requiring great strength. Hands are also capable of expressing intended emotion through touch and may also be used as weapons. Humans have feet capable of transport of the body through widely varying types of terrain. Their visual and auditory systems are certainly far superior to the Daleks. Humans have very sophisticated and powerful communication capabilities through speech — not only for expressing ideas, but also deeply and emotionally expressive as in music — quite



Fig 1 A typical Dalek/human interaction [© BBC].

unlike the limited command-only mode of speech characteristic of Daleks.

This paper is about an increasing tendency of humans to put themselves into Dalek-like worlds, where they meet and interact with others similarly engaged. Such worlds are called ‘collaborative virtual environments’ (CVEs) and also ‘shared virtual environments’ — the latter perhaps a more accurate term since an environment itself cannot be ‘collaborative’, and begs the question of whether the individuals in it can actually collaborate.

The purpose of this paper is to review some experiments that explore what happens when humans do meet one another in such Dalek-like situations. How do they interact with one another? To what extent do social conventions hold? Can people become shy or angry or generally

¹ <http://www.physics.mun.ca/~sps/whonews.html>

experience social discomfort? If the group of people so meeting have a task to do in common, does a leader emerge, and is leadership capability associated with their computational resources? This paper considers the question of what happens when real people meet other real people, but through the medium of virtual space and virtual personal embodiment.

An avatar is the virtual body inhabited by a participant in a virtual environment, who may be a real person, or entirely computer generated. There is then a subsidiary question: suppose the beings that are met by a human in a shared virtual environment are not actually human, but avatars controlled by a computer program. What happens when such real people meet other virtual people — virtual not just in their representation, but in their state of being? The same questions can be asked — do real people experience, for example, social discomfort if a virtual being does something socially unpleasant to them, or on the other hand do they experience pleasure when a virtual person is ‘nice’ to a real person?

It is likely to be the case that more and more interaction between people will take place through encounters where the shared space is entirely virtual. The people involved may be physically separated across thousands of miles. The normal capabilities of people (visual, auditory, locomotory, kinesthetic and tactile) may be grossly impaired compared to reality, and there may even be doubt about whether some of those involved are ‘real people’ or just ‘avatars’ with an automatically determined behaviour. It is important to understand how the technology influences the likely social behaviour, and the consequences of this for the types of social interaction that might develop.

This paper reports a series of experimental studies that examine these questions. The focus of the studies was that of real people meeting in a virtual setting — where the real people had never met in reality, and only interacted through the virtual medium. The people met in order to carry out a task designed by the experimenters. A further study examines the reaction of people to audiences in a virtual setting, where those audiences are completely virtual. The purpose was specifically to examine whether positive audience responses resulted in a positive effect in the experimental subjects, and similarly whether negative responses resulted in a negative effect.

In section 2 the scenario for the main group of experiments is described. Two main issues are considered — the social comfort or discomfort experienced by the participants (section 3) and the emergence of a task leader (section 4). In section 5 there is a report of a pilot study on reactions to a virtual audience. Conclusions are presented in section 6.

2. Scenario for three shared VE experiments

2.1 Organisation

Three different experiments have been carried out, each involving several groups of three people each. Different groups were used for each of the experiments. The first experiment [1] was carried out at University College, London (UCL), and involved each of the ten groups involved meeting first virtually to carry out a task, and then continuing the same task meeting face-to-face. The second experiment [2] was carried out as part of the COVEN project, where the three participants in each of the four groups involved were in widely different locations — one at UCL, another at the University of Nottingham, and the third at the Swedish Institute of Computer Science. The groups carried out the same task as in the original experiment, but these people never met physically.

The final set of experiments [3] consisted of 20 groups of three, again as part of the COVEN evaluation trials, and were located at UCL, Nottingham and at the company IIS in Athens, Greece. Again, of course, the people never met to continue the same task in reality. The common features between all three experiments are described next.

Each subject was assigned a colour (Red, Green, or Blue) as their name, and this colour also signified their appearance. The person named ‘Red’ had a red avatar, and similarly for the other two colours. No real names were ever used — the three people only referred to themselves and to others as Red, Green or Blue.

Wherever located, each subject was assigned to a ‘minder’ responsible for taking them through the various stages of the experiment. They were first introduced to the system that they would be using. This was either a desktop system or an immersive system with a head-mounted display.

The virtual environment displayed was actually a rendition of a real laboratory at UCL.

Their first task was for each individual to learn to move through the environment. Then, at a signal from the overall controller of the experiment to the various minders, each subject was given a sheet describing the task to be performed. Then again on a signal they were invited to put on earphones, and to introduce themselves to one another, only referring to themselves and to the others by their colour.

The task was to locate a room which had sheets of paper stuck around the walls. The sheets each had several words in a column, each preceded by a number. The words across all sheets with the same prefix number belonged to a 'saying'. For example, 'A critic is a man who knows the way but can't drive a car', would be split up across several sheets, but each word belonging to this sentence had a common prefix number. The task was first to figure out that this was the puzzle to be solved, and second to unscramble as many of these sayings as possible.

The total scene consisted of about 3500 polygons, which ran at a frame rate of no less than 20 Hz in stereo. The latency was approximately 120 ms.

2.2 Special features of the UCL experiment

In the first experiment, the one entirely located at UCL, the subjects carried out this task in the virtual environment for 15 minutes. They then stopped to answer questionnaires individually. Then each subject was requested to put on a waistcoat corresponding to their colour, and they were brought together to meet for real. They carried on the task for a further 15 minutes in the real laboratory on which the virtual laboratory had been modelled. (The papers were stuck around the walls in the same arrangement as they had been virtually). After this real session, the subjects again completed their questionnaires, and then met together with the experimenter for a de-briefing session.

In this first experiment the Red subject was immersed, and the Green and Blue subjects were using desktop systems².

² The Red (immersed) person was using a Silicon Graphics Onyx workstation with twin 196 MHz R10000, Infinite Reality Graphics and 64M main memory, running Irix 6.2. The tracking system had two Polhemus Fastraks, one for the HMD and another for a 5 button 3D mouse. The helmet was a Virtual Research VR4 with a resolution of 742 x 230 pixels for each eye, 170 660 colour elements and had a field-of-view 67 degrees diagonal at 85% overlap.

The Green subject used an SGI High Impact system with 200 MHz R4400 and 64 MB main mamory. The scene was shown on the full 21 inch screen display. Navigation was accomplished by using the keyboard arrow keys, with up and down arrows giving forward and back movement, and left and right keys providing rotation. All movement was on the horizontal plane of the floor.

The Blue subject used an SGI 02 running at 180 MHz on Irix 6.3, with an R5000 processor, and 32 MB main memory. The scene was shown on a full 17 inch screen display. Naviation was the same as for the SGI Impact.

The sound system used was the Robust-Audio Tool (RAT) v.3.0.23. This allows multiple users to talk over the Mbone network [4].

The virtual reality software used throughout was DIVE 3.2 from the Swedish Institute of Computer Science. A DIVE avatar was used for each of the participants, and was the same for each except for the colour.

There was a further secret instruction given to the Green subject in the virtual part only of this experiment — this was to get in the way of the Red person as much as possible — to try to always be in Red's line of vision. The purpose was to examine the degree of social discomfort generated in the group as a result of this behaviour.

2.3 Special features of the UCL-Nottingham-SICS experiment

There were four groups of three subjects each, none knowing either one other, or the purposes of the experiment.

No person was immersed — though there were differences between the machines used — this would help to find if immersion accounted for the leadership effect in the earlier experiment or whether it was more to do with speed of machine³.

As in the first experiment, none of the avatars had any expressiveness or limb movements. In the first experiment the three avatars used were identical apart from colour. In this second experiment, in order to examine the potential impact of avatar appearance, one avatar, the one at SICS, was different to the other two. The Red and Blue subjects had basic male caricature avatars. The third (Green, at SICS) was represented in a more realistic manner. It was also smaller than the other two. The SICS (Green) person would see two other avatars of the same cartoon-like appearance apart from colour; whereas each of the other two participants would see one avatar looking very realistic and the other cartoon-like. None of the subjects had any idea as to how they themselves were portrayed.

When the Red or Blue people spoke, a wave-like emanation would be emitted from their avatars. This was not the case for Green — another difference between the Green person and the other two.

Heterogeneous IP networks were used to connect the sites together and the world and bodies which were used were accessed via HTTP from SICS throughout.

2.4 Special features of the UCL-Nottingham-IIS experiment

For this third experiment the dVS/dVISE 5.0 system from Division Ltd was used to implement the scenario⁴. Machines with different capabilities were used at each of

³ At SICS an SGI 02, R5000 180 MHz with 96MB main memory was used. At Nottingham a High Impact R4400 200 MHz with 192 MB main memory was used, and at UCL a high impact R4400 CUP 200 MHz with 64MB main memory was used. All monitors were 21 inch, and the full monitor screens were used.

⁴ dVS does not supply audio communication between participants, hence the Robust-Audio Tool (RAT) v.3.0.23 was used alongside the main application.

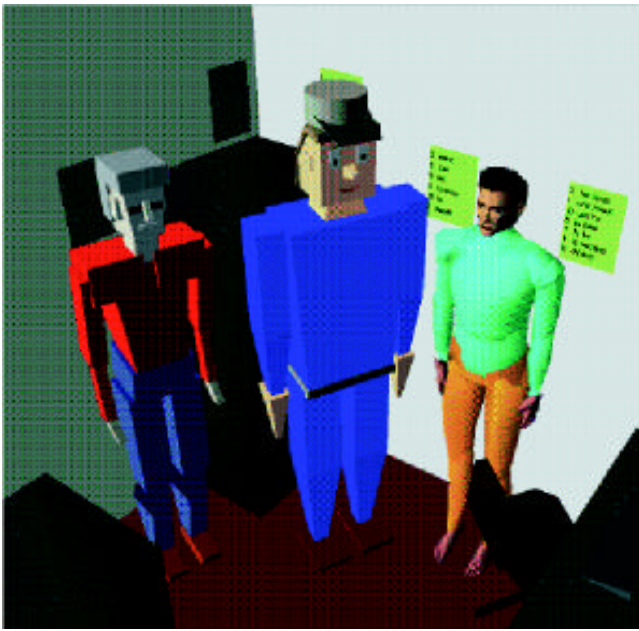


Fig 2 The three types of avatars used in the three experiments are shown in the virtual room. The Red one is the type of avatar used in the first experiment. The Green one is the more sophisticated avatar used in the second and third experiment. The Blue one was the cartoon-like one used in the second experiment.

the research sites⁵. The model ran at 20–30 Hz on all client machines.

The experiment was carried out over a network used for weekly trials between the COVEN partners. Each site was connected over ISDN to UCL. The dVS master server ran at UCL, with each client downloading the initial state from the server and then peer-to-peer connections are made between the sites for the delivery of events. An audio server ran on the same machine and sites connected using the system's point-to-point connection capability.

3. Social discomfort in shared VEs

There are two sources of result regarding social discomfort — detailed statistical analyses of the questionnaire responses, and reports of the de-briefing sessions held with the subjects at the conclusion of the experimental trial. Detailed results of the statistical analyses of the questionnaire results are available in Slater et al [1]; the main conclusions only are presented here.

⁵ The participant machine at UCL was a Silicon Graphics Onyx with twin 196 MHz R1000 processors, Infinite Reality graphics and 192 M of the main memory. The participant machine at Nottingham was a Silicon Graphics Indigo2 with a 200 MHz R4400 processor, High Impact graphics and 192 M of main memory. The participant machine at IIS was a Silicon Graphics Octane with a 195 MHz R100000 and 128 M main memory. The final machine involved was the dVS server and audio server at UCL. This was a Silicon Graphics Indigo with a 200 MHz R4400 processor, High Impact graphics and 64 M of main memory. The participant at UCL was using an immersive system employing a 2-tracker Polhemus Fastrak system. Virtual Research VR4 helmet and a 3-D mouse with 5 buttons. The other two participants were using a desktop system using the full 1280 x 1024 screen and a standard 2-D mouse with three buttons.

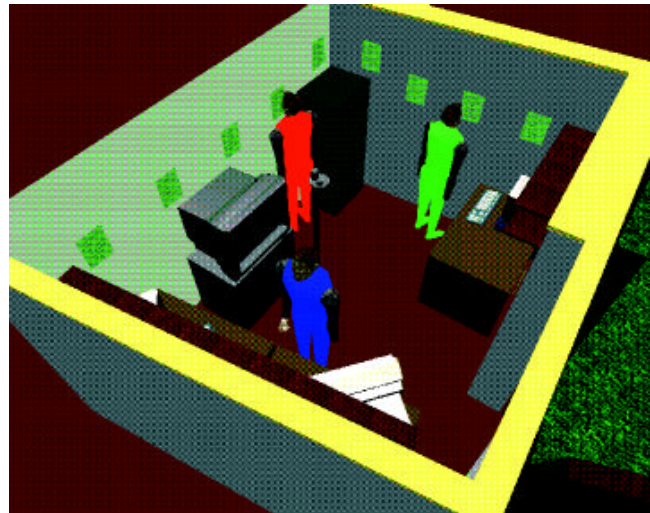


Fig 3 The three avatars in the third experiment.

3.1 Comparing virtual to real

In the first (UCL-based) experiment it was possible to compare responses given after the real and virtual sessions. Of course, it is impossible to know if the different responses were simply due to a time effect (the real sessions were always conducted after the virtual). More experiments are needed where the order of presentation would be reversed.

The questionnaire (the same one given after each of the virtual and real sessions) explored various aspects of accord between the group members, and the aspects of accord listed below had significantly different responses as between the virtual and real sessions — in each case the virtual session was experienced as the one with the greater level of social discomfort:

- enjoyment of working with others in the group (compared to a previous enjoyable group experience),
- the degree of isolation of members from the group,
- the extent of feelings of comfort with the other group members,
- the degree to which group members co-operated with one another.

When all 'accord' variables were used to construct one overall measure of group accord, there was an overwhelmingly significant difference between the virtual and real sessions, with the real session resulting in a much greater level of overall accord than the virtual.

The reason for the differences might not be solely due to the nature of a virtual compared to a real encounter. Another factor that was different between the two sessions was that in the virtual session Green was asked to 'monitor' Red, while this was not the case in the real session. However,

when the responses for the individuals are examined, there are no significant differences between Red, Green and Blue for any of the 'accord' variables.

3.2 *The impact of interference*

Although it did not show up in the questionnaire responses, there was a clear impact of the monitoring task, in the first UCL-only experiment — where Green was given the additional instruction to get in Red's way.

In three of the ten groups there was an impact of this additional task by Green. In one group Red formed the opinion that Green was being deliberately destructive. All three members of this group (Red and Green male, Blue female) had a high sense of what they described as 'paranoia' during the virtual session, and which completely disappeared when they met for real.

In another group Red did notice something different - but interpreted this as something being wrong with the avatar configurations. She said that: 'Everyone was supposed to be looking at the walls, but Green was looking at me'. In this same group, Green reported that: 'I felt I wasn't being me' and 'What on earth were they thinking of me?' She imagined that the other two were '...wondering why I am doing this.'

In another group the major impact was on Blue, who thought that Red and Green 'know where they are — I felt excluded'. In other words Blue noticed that Red and Green seemed to be close to one another most of the time, and Blue was left out of this.

One thing reported by almost all Green subjects was the difficulty of carrying out the monitoring task successfully. Green could not judge Red's field of view. There was no 'eye contact' in any meaningful sense, no 'exchange of glances'. More generally this lack of feedback about body movements and body language from the avatars was mentioned by several people.

3.3 *Experiencing avatars*

A major issue explored in the de-briefings was the relationship of the people to their avatars. The most interesting way in which this was realised was through projection — that is, individuals were respectful of the avatars of the other people, and tried to avoid carrying out actions that would cause distress or which would be impossible in real life. This process of being mindful of the avatars of others was surprising, they were taken seriously in spite of all their shortcomings.

The analysis of the post-experimental group discussion revealed a surprising degree of attachment and relationship towards the virtual bodies (avatars). Although, except by

inference, the individuals were not aware of the appearance of their own body, they seemed to generally respect the avatars of others, trying to avoid passing through them. These were very simple avatars, with limited movement and no capability for any kind of emotional expression.

The behaviours that were observed or reported included various types of embarrassment or social discomfort on accidentally walking through another person's avatar. This behaviour was variously described as rude, frightening, uncomfortable, disconcerting, or just plain 'bad'. Some group members were observed to apologise when they walked through another person.

In the second study (UCL-Nottingham-SICS), recall that two of the avatars (Red and Blue) were cartoon-like, and the third looked very realistic, although it had no more functionality than the other two. The differing avatar representations clearly had an impact on the relationships that developed. In one group the realistic avatar was experienced as scary — it looked real but was like a 'zombie' since the degree of realism raised expectations about corresponding realistic movements which, of course, did not occur. In another group, the major impact of the avatar representations was on the subject who saw the other two avatars as cartoons. This subject formed the belief that the cartoon-like avatars were not embodying real people but were 'robots', and as a result she cut down her communication with them. It was only when they laughed ('something a robot cannot do') that she believed they were real, and initiated a greater level of communications.

In another group, one of the subjects formed the opinion that the realistic avatar embodied an important person, a senior person, a managing director, someone with more privilege. This led to a further belief that this person would understand more about the system as a whole — simply because the body was more sophisticated than that of the other avatar that could be seen.

In a final group there was no reported impact of the body representations. However, the UCL-based subject, believed, because of poor audio in that session, that the other two subjects were talking in German, and possibly talking about him, and laughing at him. There was no particular reason for him to form this belief (of course, one of the subjects was in Nottingham and the other in Sweden, their common language being English). However, this became a very powerful belief, and, he reported, had a similar impact to what he would have felt in real life. 'You become a minority,' was his closing remark.

In all three experiments aspects of avatar functionality and behaviour were reported as hindering the social collaboration. The lack of emotional expressiveness (except through voice) was overwhelmingly mentioned. The lack of eye-contact, body language, and even the ability to point at

a reference object (one of the sheets of paper on the wall) were important drawbacks. Subjects seated at the screens could be seen to frequently point with their real hand at an item on the virtual wall, while speaking about it to the others. This seems to be an irrepressible behaviour, even though the people are intellectually aware that their real gesture cannot possibly be seen by the other people in the group. Perhaps this is a significant argument for an immersive system, where participants really do point with their real hands, which can be seen by the others involved. The impact of immersion on a particular aspect of social relationships is discussed in the next section.

4. Emergence of a leader in shared VEs

The nature of the task required collaboration for a successful resolution. Papers were distributed around the walls of a room, and the subjects were unable to take notes (even in the real meeting). Ideally each subject would stand at a different part of the room, and relate the words and numbers they could see to the group as a whole. The spatial arrangement of the group and the method of working would often come about through one person suggesting ideas to the others. This person would be perceived as leader and also do most of the talking. In the questionnaire used for each experiment, each person was asked to rate the three people (including a self-assessment) on the extent to which they emerged as leader, and the extent to which they did the most talking. These two scores (leadership and talkativeness) were very highly correlated for the two experiments with enough subjects (the first, 30 people, and the third, 60 people) to make the correlations meaningful. For example, in the third experiment, the correlation was over 0.9.

In the first experiment one person was immersed and the other two were on desktop systems. Whether averaging the leadership and talkativeness scores, or counting the number of times that the immersed person was chosen as leader, or chosen as the most talkative, there was a very high statistically significant bias towards the immersed person. This result was not the case for that person when the groups met for real. It should be noted at this point that the subjects did not know about the systems used by their colleagues, and would have had no idea that one of them was immersed.

In the second experiment none were immersed, although there were differences in speed and performance of the various machines involved. However, no particular leadership pattern emerged.

In the third experiment again one person was immersed. The result is more complex though. On the raw data for leadership and talkativeness counts and scores there was no particular pattern in favour of the immersed person. The only conclusion from the raw data in this regard was that the person in Greece was almost never rated as leader. This was

probably because the network connection was 5 to 10 times slower than that between London and Nottingham, and, of course, there was also the question of language difference.

A detailed statistical analysis, however, revealed a more interesting picture. In the third experiment, the subjects answered a questionnaire concerned with their general social anxiety, called the Interaction Anxiousness Scale [5]. Generally a higher score means a higher level of general social anxiety, and average scores for the general population, and also for those seeking counselling or psychiatric help are known. The hypothesis was that there would be a negative correlation between social anxiety and degree of leadership. This was indeed the case, with higher levels of social anxiety associated with a lower leadership rating. Now when social anxiety was factored out of the relationship between immersion and leadership score, an interesting result was found — for males, there was a clearly greater leadership score for the immersed subjects than for the non-immersed ones; for females, the immersed and non-immersed subjects had almost the same leadership rating. The reason for this is unknown; possible reasons include the general discomfort and weight of the HMD, and the fact that the use of the HMD system is certainly more intimidating than the familiar activity of just looking at a screen. Alternatively, it could just be a chance result; clearly more studies are needed to reach a firm conclusion about this issue. Certainly, being able to compare behaviour in the real and the virtual environment is a crucial part of understanding, and this was not possible in the second and third experiments.

5. Responses to a virtual audience

This section reports a different experiment, one focused on interactions between real people and virtual humans represented by avatars within the VE. Its application area is the use of virtual environments for the treatment of social phobias.

One of the major potential applications of virtual reality is in the area of psychotherapy, and there are already some results in the literature on this [6, 7]. Most of the work has been concerned with fears and phobias of 'things' or situations involving things — spiders, bridges, airplanes, heights. A more interesting and challenging application is in social phobias — shyness, various types of social anxiety, including fear of public speaking. These phobias can cause enormous damage in people's lives, and are difficult and costly to treat, because to confront these fears other people must be confronted. This is both difficult to organise, and is the very heart of the problem itself. It is the same sort of dilemma faced in the treatment of agoraphobia — the client has to go out into the open in order to get to the therapist in order to receive help, and yet it is the 'going out' which is the whole problem.

The area of therapy and in particular treatment of social phobia is also of theoretical interest. Virtual reality offers the possibility of 'presence' in a virtual place and co-presence with other people located at remote spaces — and there has been a great deal of research in this area (a review can be found in Slater and Wilbur [8]). Therapy for social phobia cannot work without a strong sense of presence-in-a-place (e.g. the seminar room for fear of public speaking therapy) and with other people (the virtual audience). Maximising both types of presence is of great practical utility in this type of application.

For a Masters of Research project at UCL, carried out by David-Paul Pertaub [9] a preliminary investigation has been carried out into the feasibility of virtual reality based therapy for fear of public speaking. Classical exposure therapy gradually introduces the client to the feared 'other' — in this case the experience of speaking in front of an audience. Yet, clearly it is complicated and expensive to find real audiences who would be willing to perform the role of becoming an audience for a client — and there are potentially millions of such clients [10].

In this study speakers were recruited on the UCL campus by asking for volunteers who wished to rehearse a short talk of about five minutes, in a 'safe' setting provided by a virtual audience. The scenario was, for example, a job interview, or an examination viva where the person is asked: 'Please give a brief description of your current work.' The subjects were free to choose any topic of their choice, and to speak without notes or other prompts.

The purpose of the study was not to provide therapy, but only to see whether the subjects would respond with negative affect to negative audience responses and with positive affect to positive responses. There were three types of programmed audience response: negative, where the virtual audience members independently looked away from the speaker, avoiding 'eye contact', fidgeted, fell asleep, shook their heads, had bored facial expressions, and walked out of the room. This was to accompanying sounds of occasional yawns and moans. The positive reactions consisted of the audience gazing at the speaker, nodding, smiling, spontaneous clapping, and generally behaving in an interested way. A mixed response always started off with mainly negative responses, and gradually turned to positive responses, and ended with a standing ovation. Subjects gave their talk three times — firstly experiencing a negative or positive response, then the other way around, and finally always the mixed response.

Data was collected by questionnaire on a number of variables, including assessments of co-presence with the virtual audience, self-rating of their performance, ratings of the interest of the audience in what they had to say, and standard social anxiety and fear of public speaking questionnaires.

To date a pilot study has been completed, with 10 subjects providing full data. The results show, as would be expected, that self-rated performance generally increases with time, i.e. speakers tend to rate their performance as improving with each successive trial, on the average. However, independently of time, the speakers' self-rated performance, measured across a number of factors, is positively associated with their rating of the friendliness and mood of the audience and interest of the audience in what they had to say. Now this seems a rather trite result — clearly, anyone would rate their own performance better when the audience seems interested and friendly than when the audience is not interested and hostile. But recall — these are avatars! There are no real people there. When some programmed images seem 'friendly' and interested, the ratings of the speakers was higher, and strongly so, than when the response was experienced as not interested and hostile.

This result was not unexpected, since it is in line with other research [11] suggesting that people respond to media as other social actors. However, to see this in practice in such an experiment was truly quite stunning. There is so little that can be retained by questionnaires — the whole body language of some of the speakers was quite different after experiencing a positive session than a negative session. One speaker, on his second time around after previously experiencing a negative session, apologised to the audience saying: 'I am sorry I bored you last time, I will try to do better this time.' Another, on seeing the audience walk out on him said: 'I can see that you're not interested in my subject.' Overall it is the responses that we could not capture through questionnaires that were the most significant, rather than the pale reflection that questionnaires can give of such an experientially rich situation.

This study is continuing, with further refinements to the software describing and controlling the avatars, and different presentation media such as large display screens. It clearly has applications for training as well as psychotherapy.

6. Conclusions

This paper has reviewed some experimental work on the responses of people to virtual environments that involve other social actors. The results suggest that people do respond strongly to the representations of other people, as people. Further, however, that the accord generated within a group may not be as strong as it would be if the same people met for real. There are very significant impediments to social interaction caused by the lack of the ability for emotional and gestural behaviour and response. There is evidence to suggest that immersion enhances the capability for leadership, though this is a complex issue and the matter is far from settled. Finally, there is evidence to suggest that

significant emotional responses can be obtained even when the human participants know that there are no real people there, just computer controlled avatars that have some basic human-like behaviours. In this experiment, people rated their own performance according to the perceived interest of a group of avatars. In terms of the wide area international trials, perhaps the most telling and positive result, overlooked in the context of carrying out the experiment, is that the experiments themselves were managed using the shared virtual environments systems (DIVE or dVS, both part of the COVEN platform). These experiments were logistically complex, and yet were managed in real time by the experimenters using the shared VE system.

The Daleks are a nasty race, interested only in commanding, conquering and destroying. It is possible that they behave in this way because of their (in comparison to humans) warped perception of the real world. It is extremely likely that in years to come a large proportion of the world's business and social activity will be carried out in extensive shared VEs. Therefore we have to make sure that we do not evolve a breed of humans who behave like Daleks because of their similarly warped and limited perception and potential for action in the virtual world. Much research of this kind is essential, in order to understand what happens when people meet other virtually real people in virtual places.

Acknowledgements

Most of the work described in this paper is funded through the ACTS project COVEN — Collaborative Virtual Environments. The COVEN consortium consists of twelve partners: Arax Ltd (UK), Division Ltd (UK), EPFL (Switzerland), IIS Ltd (Greece), KPN Research (The Netherlands), Lancaster University (UK), SICS (Sweden), Thomson-CSF LCR (France — co-ordinator), TNO FEL (The Netherlands), University College London (UK), University of Geneva (Switzerland), and the University of Nottingham (UK).

References

- 1 Slater M, Sadagic A, Usoh M and Schroeder R: 'Small group behaviour in a virtual and real environment: a comparative study', accepted for publication in *Presence: Teleoperators and Virtual Environments* — a summary can be found on <http://www.cs.ucl.ac.uk/staff/m.slater/BTWorkshop/Inhabit/> (1998)
- 2 Tromp J G, Steed A, Frecon E, Bullock A, Sadagic A and Slater M: 'Collaborative virtual environments: experiments on small group behaviour in the COVEN Project', *IEEE CG&A* (November 1998).
- 3 Steed A, Slater M, Sadagic A, Bullock A and Tromp J: 'Leadership and collaboration in shared virtual environments', submitted for publication (1998).
- 4 Hardman V, Sasse M A and Kouvelas I: 'Successful Multiparty Audio Communication over the Internet', *Communications of the ACM*, **41**, No 5, pp 74—80 (1998).
- 5 Leary M: 'Social anxiousness: the construct and its measurement', *Journal of Personality Assessment*, **47**, pp 66—75 (1983).
- 6 Rothbaum B, Hodges L, Kooper R, Opdyke D, Willford J and North M: 'Effectiveness of computer generated (virtual reality) graded exposure in the treatment of acrophobia', *American Journal of Psychiatry*, **152**, No 4, pp 626—628 (1995).
- 7 Hodges L F, Rothbaum B O, Watson B, Kessler G D and Opdyke D: 'A virtual airplane for fear of flying therapy, *IEEE 1996 Virtual Reality Annual International Symposium*, pp 86—94 (1996).
- 8 Slater M and Wilbur S: 'A framework for immersive virtual environments (five): speculations on the role of presence in virtual environments', *Presence: Teleoperators and Virtual Environments*, **6**, No 6 pp 60—616, MIT Press (1997).
- 9 Pertaub D P: 'Virtual exposure therapy for public speech anxiety', Master of Research Thesis, Department of Computer Science, University College London (1998).
- 10 Schneier F and Welkowitz L: 'The hidden face of shyness', *Understanding and Overcoming Social Anxiety*, Avon Books, New York (1996).
- 11 Reeves B and Nass C: 'The media equation: how people treat computers, television, and new media like real people and places', CSLI Publications and the Press Syndicate for the University of Cambridge (1998).



Mel Slater is Professor of Virtual Environments at University College, London. His main research interests are real-time global illumination in computer graphics, and shared environments.

He is a Principal Investigator on a number of projects — the ACTS Collaborative Virtual Environments project, the Digital Virtual Centre of Excellence (Digital-VCE) project on Virtual Rehearsal for Actors, and the Foresight VR Centre for the Built Environment.